

# Week 1 - Foundations of Cryptography

Merve Hatun Özcan

September 2025

## Foundations and Essential Tools of Cryptography

### Session 1 – Foundations of Cryptography

Cryptography is securing communication and protocols against adversaries. At its core, it provides the building blocks called *primitives* that allow us to design secure systems.

#### Ciphers

A cipher transforms readable data plaintext into unreadable noise ciphertext using a secret key. In **symmetric encryption**, the same key is used to both encrypt and decrypt messages. Security rests not on hiding the algorithm but on keeping the key secret, a principle known as **Kerckhoffs’ principle**.

Kerckhoffs’s Principle, formulated in 1883, outlines the fundamental requirements for a secure cryptographic system. It states that a system should remain secure even if its design becomes public, relying only on the secrecy of the key. The key must be easy to remember and change without written notes, and the system should be practical, portable, and operable by a single person without undue complexity. Additionally, it must be resilient in real-world use, applicable to communication technologies of the time, and simple enough that users can operate it without stress or the need for extensive rules. These principles emphasize usability and robustness as essential aspects of security, not just mathematical strength.

#### Number Theory Basics

Concepts such as *prime numbers*, the *greatest common divisor (GCD)*, and *modular arithmetic* are fundamental to cryptography. Modular arithmetic (computations “wrapping around” a modulus) enables the construction of hard problems, such as the discrete logarithm problem, which form the basis of secure protocols.

## Session 2 – Essential Tools of Modern Cryptography

Modern cryptography relies on a toolkit of primitives beyond ciphers.

### Hash Functions

A hash function maps input data of any length to a fixed-size output (*digest*). Good hash functions are deterministic, collision-resistant, and hard to invert. They are widely used in commitments, integrity checks, and password storage.

### Symmetric Cryptography

Efficient algorithms such as AES protect confidentiality of large amounts of data. However, they require secure key distribution mechanisms.

### Key Exchange

Key exchange protocols establish a shared key over insecure channels. The classic example is **Diffie–Hellman (DH)**, which allows two parties to agree on a secret without prior contact.

### Asymmetric Cryptography

This paradigm uses **public/private key pairs**. Anyone can encrypt with the public key, but only the private key owner can decrypt. It also enables digital signatures for authentication.

### Public–Private Key Diffie–Hellman

An extended version of DH that uses long-term public/private keys, ensuring stronger guarantees and enabling secure protocols such as TLS. The elliptic-curve variant (ECDH) is widely used today.

Together, these tools form the essential toolkit of modern cryptography: ciphers for confidentiality, hashes for integrity, symmetric systems for speed, and asymmetric systems for secure key distribution and authentication.